Automatidata Project: Predictive Modeling of New York City Taxi Fares

Linear Regression Pipeline, Data Cleaning, and Cost Factor Analysis

Overview

Analysis of over 20,000 New York taxi records to accurately predict the total trip cost (fare_amount).

Problem

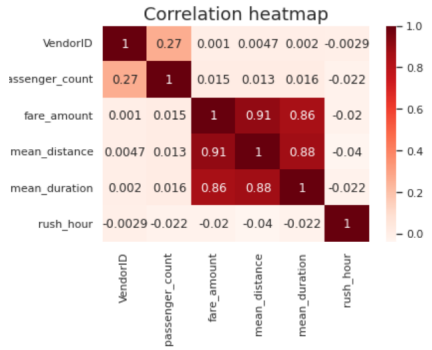
Data containing outliers, negative rates, and zero-value records, which prevented reliable price estimation.

Solution

Pipeline in Python: cleaning using the Interquartile Range (IQR), feature engineering (peak hours and zonal averages), and a Linear Regression model.

Details

Findings: Strong correlation of distance (mean_distance, $r=0.91$) and duration (mean_duration, $r=0.86$) with the fare. Model: Application of StandardScaler and One-Hot Encoding to evaluate coefficients.



Next Steps

Integrate the model into the operational platform to provide real-time fare estimates to users.